Supplemental Materials

Train–Test Leakage in Pothole Detection Benchmarks: Evidence from a Video-Disjoint Re-Split of BharatPotHole

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Table S1 Leakage criteria in the published test set (A-test, n = 662)

Criterion	Images	%
Confirmed near-duplicate in training ($d_H \leq 4$, or $5 \leq d_H \leq 8$ with SSIM ≥ 0.80)	259	39.1
Shared source frame with a training image	242	36.6
$d_H \leq 4$ to the nearest training image	257	38.8
$5 \leq d_H \leq 8$	76	11.5
$5 \leq d_H \leq 8$ and confirmed by SSIM	2	0.3
Confirmed near-duplicate and shared source frame	242	36.6
Confirmed near-duplicate only	17	2.6
Shared source frame only	0	0.0
Leaked (either criterion)	259	39.1
$d_H \leq 4$ but not leaked	0	0.0
Leaked with $d_H > 4$	2	0.3
From a video not used for training	4	0.6

d_H: Hamming distance between 64-bit perceptual hashes. Leaked + B-clean = 259 + 403 = 662 images
B-strict (337 images) also excludes 66 positive images with no near-duplicate in training (minimum $d_H = 8$) but with confirmed near-duplicates in the validation set (38) or within the test set (48)

Table S2 Sensitivity of the A-test–B-clean gap to the near-duplicate definition (YOLO26n, Protocol A, positive images)

Reference subset	Positive images	Videos	AP_{50}	ΔAP_{50} [95% CI]	Relative gap (%)
$d_H > 2$, no shared source frame	127	36	0.465	0.154 [0.093, 0.238]	33.1
$d_H > 4$, no shared source frame	125	36	0.467	0.152 [0.090, 0.234]	32.6
$d_H > 6$, no shared source frame	122	34	0.460	0.159 [0.099, 0.245]	34.6
$d_H > 8$, no shared source frame	99	31	0.448	0.171 [0.100, 0.304]	38.2
B-clean (main analysis)	125	36	0.467	0.152 [0.090, 0.234]	32.6
B-strict	59	29	0.445	0.174 [0.098, 0.283]	39.0

AP_{50} of A-test (positive images) = 0.619. Paired bootstrap over videos, 1,000 resamples, percentile intervals
B-strict with 5,000 resamples: 95% CI 0.092–0.274; share of resamples with $\Delta AP_{50} \leq 0$ = 0.02%

Table S3 Composition of training and test sets (cleaned labels)

Set	Images	Positive images	Boxes	Boxes per image	Small boxes (%)	Median box (px)	Videos	Videos with positives
Protocol A training	5,067	3,062	8,777	1.73	60.5	26.8	153	–
Protocol C training	2,927	934	2,836	0.97	56.3	28.9	136	–
A-test	662	371	1,148	1.73	51.8	30.9	93	53
Leaked	259	246	643	2.48	58.9	27.6	50	40

B-clean	403	125	505	1.25	–	–	79	36
B-strict	337	59	201	0.6	41.8	35.3	72	29
C-test	425	134	381	0.9	56.4	29.6	10	9

Small boxes: square root of box area < 32 px. Unique positive source frames: 1,157 in Protocol A training and 934 in Protocol C training
C-test: the video with most positive images contributes 57 of 134

Table S4 Pools used in the controlled contamination test

Pool	Images	Boxes	Boxes per image	Small boxes (%)	Median box (px)
C-test positive images (original)	134	381	2.84	56.4	29.6
Export copies of training frames	726	2,235	3.08	57.4	28.4
Export copies of C-test frames (unseen)	112	322	2.88	54.3	30.3

One copy per source frame. AP_{50} of all 112 unseen copies without substitution: 0.193 (YOLO26n) and 0.189 (YOLO26s)

Supplemental Figures S1–S6

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Fig. S1 Published test images (left of each pair) and their nearest training image (right). Green boxes: ground truth. d_H: Hamming distance between perceptual hashes; SSIM on 128 × 128 grey thumbnails



Fig. S2 All released files that share one source frame (four export copies), with the subset of the published split each was assigned to. Green boxes: ground truth



Fig. S3 YOLO26n trained on the published split (Protocol A) on test images with a near-duplicate in training (Leaked, top) and without (B-clean, bottom). Green: ground truth; orange dashed: predictions with confidence ≥ 0.25



Fig. S4 Controlled contamination test: YOLO26n trained on the video-disjoint split (Protocol C) on C-test images (top), export copies of unseen C-test frames (middle) and export copies of its own training frames (bottom). Green: ground truth; orange dashed: predictions with confidence ≥ 0.25



Fig. S5 Pairs of near-identical unique frames ($d_H \leq 4$) whose annotations differ by two or more boxes. Green boxes: ground truth



Fig. S6 Random positive images from BharatPotHole (top) and from the RDD2022-India pothole subset (bottom). Green boxes: ground truth (class D40 for RDD2022)